CS 2601 Linear and Convex Optimization

5. Convex optimization problems (part 2)

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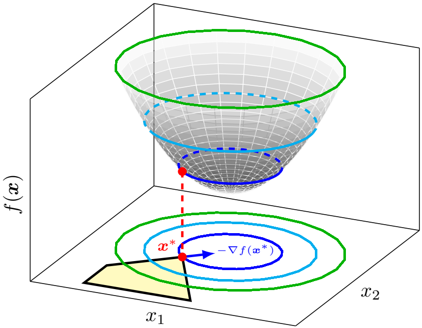
Outline

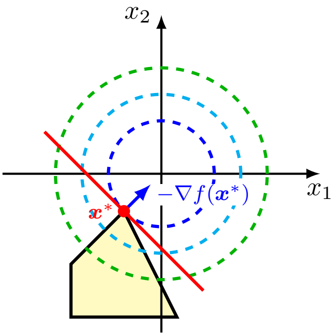
- Quadratic program and quadratically constrained QP
- Geometric program

Quadratic program (QP)

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & \frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} + \boldsymbol{c}^T\boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{B}\boldsymbol{x} \leq \boldsymbol{d} \\ & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \end{aligned}$$

QP is convex iff $\boldsymbol{Q} \succeq \boldsymbol{O}$. Reduces to LP if $\boldsymbol{Q} = \boldsymbol{O}$.

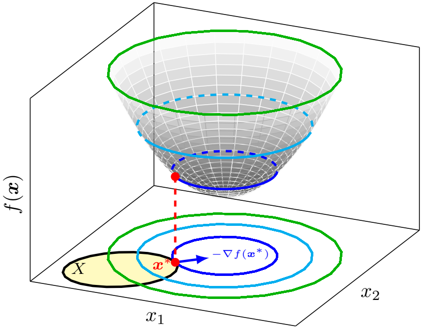


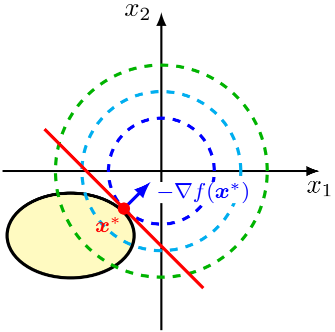


Quadratically constrained quadratic program (QCQP)

$$
\begin{aligned}
\min_{\boldsymbol{x}} \quad & \frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} + \boldsymbol{c}^T\boldsymbol{x} \\
\text{s.t.} \quad & \frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}_i\boldsymbol{x} + \boldsymbol{c}_i^T\boldsymbol{x} + \boldsymbol{d}_i \leq 0, \quad i = 1, 2, \ldots, m \\
& \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}
\end{aligned}
$$

QCQP is convex if $\boldsymbol{Q} \succeq \boldsymbol{O}$ and $\boldsymbol{Q}_i \succeq \boldsymbol{O}$, $\forall i$. Reduces to QP if $\boldsymbol{Q}_i = \boldsymbol{O}$, $\forall i$.





Example: Linear least squares regression

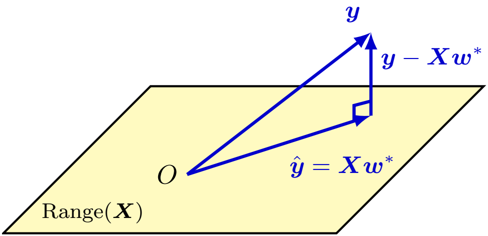
Given $\boldsymbol{y} \in \mathbb{R}^n$, $\boldsymbol{X} \in \mathbb{R}^{n \times p}$, find $\boldsymbol{w} \in \mathbb{R}^p$ s.t.

$$\min_{\boldsymbol{w}} \ \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2$$

- convex QP with objective

$$f(\boldsymbol{w}) = \boldsymbol{w}^T \boldsymbol{X}^T \boldsymbol{X} \boldsymbol{w} - 2\boldsymbol{y}^T \boldsymbol{X} \boldsymbol{w} + \boldsymbol{y}^T \boldsymbol{y}$$

Geometrically, we are looking for the orthogonal projection $\hat{\boldsymbol{y}}$ of $\boldsymbol{y}$ onto $\mathrm{Range}(\boldsymbol{X})$, the column space of $\boldsymbol{X}$. Does the solution always exist?



Example: Linear least squares regression (cont'd)

By the first-order optimality condition, $\boldsymbol{w}^*$ is optimal iff

$$\nabla f(\boldsymbol{w}^*) = \boldsymbol{0}$$

i.e. $\boldsymbol{w}^*$ is a solution of the normal equation,

$$\boldsymbol{X}^T(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}) = \boldsymbol{0} \iff \boldsymbol{X}^T\boldsymbol{X}\boldsymbol{w} = \boldsymbol{X}^T\boldsymbol{y}$$

Note. $\boldsymbol{X}^T(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}^*) = \boldsymbol{0}$ means precisely $\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}^* \perp \mathrm{Range}(\boldsymbol{X})$.

Case I. $\boldsymbol{X}$ has full column rank, i.e. $\mathrm{rank}\,\boldsymbol{X} = p$

- $\boldsymbol{X}^T\boldsymbol{X} \succ \boldsymbol{O}$
- unique solution

$$\boldsymbol{w}^* = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}$$

Case II. $\boldsymbol{X}$ does not have full column rank, i.e. $\mathrm{rank}\,\boldsymbol{X} < p$.

- $\boldsymbol{X}^T\boldsymbol{X}$ is singular
- infinitely many solutions

Example: Linear least squares regression (cont'd)

Example. Solve

$$\min_{\boldsymbol{w}} \ \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2$$

with

$$\boldsymbol{X} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}.$$

Solution. The normal equation is

$$\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{w} = \boldsymbol{X}^T\boldsymbol{y}$$

with

$$\boldsymbol{X}^T\boldsymbol{X} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{X}^T\boldsymbol{y} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

Since $\boldsymbol{X}$ has full column rank,

$$\boldsymbol{w}^* = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y} = \begin{bmatrix} 1.5 \\ 2 \end{bmatrix}$$

Example: Linear least squares regression (cont'd)

Example. Solve

$$\min_{\boldsymbol{w}} \ \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2$$

with

$$\boldsymbol{X} = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}.$$

Solution. Note $\operatorname{rank} \boldsymbol{X} = 2 < 3$. The normal equation is

$$\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{w} = \boldsymbol{X}^T\boldsymbol{y}$$

where

$$\boldsymbol{X}^T\boldsymbol{X} = \begin{bmatrix} 4 & 0 & 4 \\ 0 & 1 & -1 \\ 4 & -1 & 5 \end{bmatrix}, \quad \boldsymbol{X}^T\boldsymbol{y} = \begin{bmatrix} 6 \\ 2 \\ 4 \end{bmatrix}$$

There are infinitely many solutions given by

$$\boldsymbol{w} = (1.5, 2, 0) + \alpha(-1, 1, 1), \quad \alpha \in \mathbb{R}.$$

General unconstrained QP

Minimize quadratic function with $\boldsymbol{Q} \in \mathbb{R}^{n\times n}$ s.t. $\boldsymbol{Q} \succeq \boldsymbol{O}$,

$$\min_{\boldsymbol{x}} f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} + \boldsymbol{b}^T\boldsymbol{x} + c$$

By the first-order condition, $\boldsymbol{x}^*$ is optimal iff it is a solution to

$$\nabla f(\boldsymbol{x}) = \boldsymbol{Q}\boldsymbol{x} + \boldsymbol{b} = \boldsymbol{0}$$

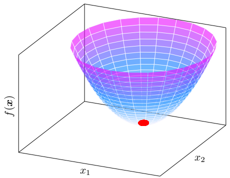
Case I. $\boldsymbol{Q} \succ \boldsymbol{O}$: unique solution $\boldsymbol{x}^* = -\boldsymbol{Q}^{-1}\boldsymbol{b}$

Case II. $\det \boldsymbol{Q} = 0$ and $\boldsymbol{b} \in \mathrm{Range}(\boldsymbol{Q})$[1]: infinitely many solutions (why?)

Case III. $\det \boldsymbol{Q} = 0$ and $\boldsymbol{b} \notin \mathrm{Range}(\boldsymbol{Q})$: no solution, and $f^* = -\infty$.

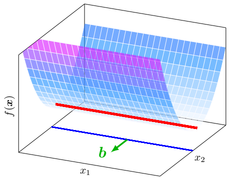
[1] This is equivalent to $-\boldsymbol{b} \in \mathrm{Range}(\boldsymbol{Q})$, the latter being just another way to say $\boldsymbol{Q}\boldsymbol{x} + \boldsymbol{b} = \boldsymbol{0}$ has a solution.

General unconstrained QP (cont’d)



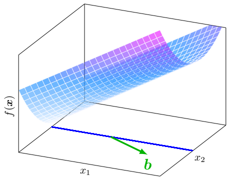
$\boldsymbol{Q} \succ \boldsymbol{O}$

unique solution



$\boldsymbol{b} \perp \mathrm{Null}(\boldsymbol{Q}) \neq \{\boldsymbol{0}\}$

infinitely many solutions



$\boldsymbol{b} \not\perp \mathrm{Null}(\boldsymbol{Q}) \neq \{\boldsymbol{0}\}$

no solution

To understand why $f^* = -\infty$ in case III, recall $\mathrm{Null}(\boldsymbol{A})^\perp = \mathrm{Range}(\boldsymbol{A}^T)$. Since $\boldsymbol{Q}^T = \boldsymbol{Q}$,

$$\boldsymbol{b} \in \mathrm{Range}(\boldsymbol{Q}) = \mathrm{Range}(\boldsymbol{Q}^T) \iff \boldsymbol{b} \in \mathrm{Null}(\boldsymbol{Q})^\perp \iff \boldsymbol{b} \perp \mathrm{Null}(\boldsymbol{Q})$$

Since $\boldsymbol{b} \not\perp \mathrm{Null}(\boldsymbol{Q})$, there exists a nonzero $\boldsymbol{x}_0 \in \mathrm{Null}(\boldsymbol{Q})$ s.t. $\boldsymbol{b}^T\boldsymbol{x}_0 \neq 0$,

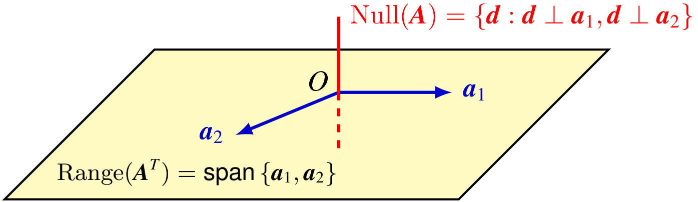
$$f(t\boldsymbol{x}_0) = t\boldsymbol{b}^T\boldsymbol{x}_0 + c$$

Letting $t \to -\infty$ if $\boldsymbol{b}^T\boldsymbol{x}_0 > 0$ and $t \to +\infty$ if $\boldsymbol{b}^T\boldsymbol{x}_0 < 0$, $f(t\boldsymbol{x}_0) \to -\infty$.

Appendix

Lemma. $\mathrm{Null}(\boldsymbol{A})^{\perp} = \mathrm{Range}(\boldsymbol{A}^T)$, where $\mathrm{Null}(\boldsymbol{A})^{\perp}$ is the orthogonal complement of $\mathrm{Null}(\boldsymbol{A})$, i.e.

$$\boldsymbol{x} \in \mathrm{Null}(\boldsymbol{A})^{\perp} \iff \boldsymbol{x} \perp \boldsymbol{d}, \quad \forall \boldsymbol{d} \in \mathrm{Null}(\boldsymbol{A})$$



Proof. Show $\mathrm{Range}(\boldsymbol{A}^T) \subset \mathrm{Null}(\boldsymbol{A})^{\perp}$ is a subspace with the same dimension, so $\mathrm{Range}(\boldsymbol{A}^T) = \mathrm{Null}(\boldsymbol{A})^{\perp}$.

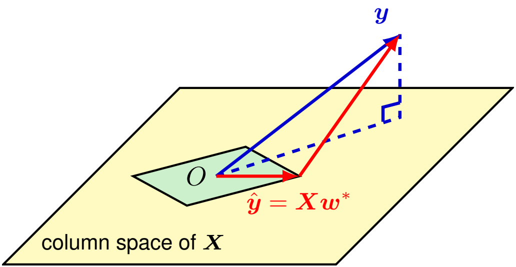
- $\boldsymbol{x} \in \mathrm{Range}(\boldsymbol{A}^T) \implies \boldsymbol{x} = \boldsymbol{A}^T\boldsymbol{z}$ for some $\boldsymbol{z}$
- $\forall \boldsymbol{d} \in \mathrm{Null}(\boldsymbol{A})$, $\boldsymbol{x}^T\boldsymbol{d} = \boldsymbol{z}^T\boldsymbol{A}\boldsymbol{d} = \boldsymbol{z}^T\boldsymbol{0} = 0$, i.e. $\boldsymbol{x} \perp \boldsymbol{d}$, so $\boldsymbol{x} \in \mathrm{Null}(\boldsymbol{A})^{\perp}$.
- $\dim \mathrm{Range}(\boldsymbol{A}^T) = \mathrm{rank}\, \boldsymbol{A} = n - \dim \mathrm{Null}(\boldsymbol{A}) = \dim \mathrm{Null}(\boldsymbol{A})^{\perp}$

Example: Lasso

Lasso (Least Absolute Shrinkage and Selection Operator)

Given $\boldsymbol{y} \in \mathbb{R}^n$, $\boldsymbol{X} \in \mathbb{R}^{n \times p}$, $t > 0$,

$$\begin{array}{ll} \min\limits_{\boldsymbol{w}} & \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2 \\ \mathrm{s.\,t.} & \|\boldsymbol{w}\|_1 \leq t \end{array}$$

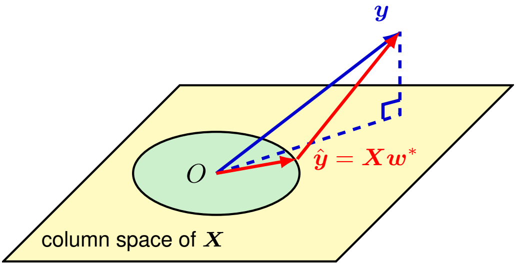


- convex problem? yes
- QP? no, but can be converted to QP
- optimal solution exists? yes
  - compact feasible set
- optimal solution unique?
  - yes if $n \geq p$ and $\boldsymbol{X}$ has full column rank ($\boldsymbol{X}^T\boldsymbol{X} \succ \boldsymbol{O}$, strictly convex)
  - no in general, e.g. $p > n$ and t is large enough for unconstrained optima to be feasible

Example: Ridge regression

Given $\boldsymbol{y} \in \mathbb{R}^n$, $\boldsymbol{X} \in \mathbb{R}^{n \times p}$, $t > 0$,

$$\begin{array}{ll} \min\limits_{\boldsymbol{w}} & \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2 \\ \mathrm{s.\,t.} & \|\boldsymbol{w}\|_2^2 \le t \end{array}$$



- convex problem? yes
- QCQP? yes

- optimal solution exists? yes
  - compact feasible set

- optimal solution unique?
  - yes if $n \ge p$ and $\boldsymbol{X}$ has full column rank ($\boldsymbol{X}^T\boldsymbol{X} \succ \boldsymbol{O}$, strictly convex)
  - no in general

Example: SVM

Linearly separable case

$$\begin{aligned} \min_{\boldsymbol{w},b} \quad & \frac{1}{2}\|\boldsymbol{w}\|^2 \\ \text{s.t.} \quad & y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b) \geq 1, \quad i = 1, 2, \ldots, m \end{aligned}$$

Soft margin SVM

$$\begin{aligned} \min_{\boldsymbol{w},b,\boldsymbol{\xi}} \quad & \frac{1}{2}\|\boldsymbol{w}\|_2^2 + C\sum_{i=1}^{m} \xi_i \\ \text{s.t.} \quad & y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b) \geq 1 - \xi_i, \quad i = 1, 2, \ldots, m \\ & \boldsymbol{\xi} \geq \mathbf{0} \end{aligned}$$

Equivalent unconstrained form

$$\min_{\boldsymbol{w},b} \frac{1}{2}\|\boldsymbol{w}\|_2^2 + C\sum_{i=1}^{n}(1 - y_i b - y_i \boldsymbol{x}_i^T \boldsymbol{w})^+$$

Outline

- Quadratic program and quadratically constrained QP
- Geometric program

Geometric program

A monomial is a function $f : \mathbb{R}^n_{++} = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{x} > \mathbf{0}\} \to \mathbb{R}$ of the form

$$f(\boldsymbol{x}) = \gamma x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}$$

for $\gamma > 0$, $a_1, \ldots, a_n \in \mathbb{R}$. A posynomial is a sum of monomials,

$$f(\boldsymbol{x}) = \sum_{k=1}^{p} \gamma_k x_1^{a_{k1}} x_2^{a_{k2}} \cdots x_n^{a_{kn}}$$

A geometric program (GP) is an optimization problem of the form

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{s. t.} \quad & g_i(\boldsymbol{x}) \leq 1, \quad i = 1, \ldots, m \\ & h_j(\boldsymbol{x}) = 1, \quad j = 1, \ldots, r \end{aligned}$$

where $f, g_i, i = 1, \ldots, m$ are posynomials and $h_j, j = 1, \ldots, r$ are monomials. The constraint $\boldsymbol{x} > \mathbf{0}$ is implicit.

Geometric program (cont'd)

GP is nonconvex (why?)

$$\begin{aligned}
\min_{\boldsymbol{x}} \quad & \sum_{k=1}^{p_0} \gamma_{0k} x_1^{a_{0k1}} x_2^{a_{0k2}} \cdots x_n^{a_{0kn}} \\
\text{s.t.} \quad & \sum_{k=1}^{p_i} \gamma_{ik} x_1^{a_{ik1}} x_2^{a_{ik2}} \cdots x_n^{a_{ikn}} \leq 1, \quad i = 1, \ldots, m \\
& \eta_j x_1^{c_{j1}} x_2^{c_{j2}} \cdots x_n^{c_{jn}} = 1, \quad j = 1, \ldots, r
\end{aligned}$$

By $y_i = \log x_i$, $b_{ik} = \log \gamma_{ik}$, $d_j = \log \eta_j$, GP can be formulated as

$$\begin{aligned}
\min_{\boldsymbol{y}} \quad & \log \left(\sum_{k=1}^{p_0} e^{\boldsymbol{a}_{0k}^T \boldsymbol{y} + b_{0k}} \right) \\
\text{s.t.} \quad & \log \left(\sum_{k=1}^{p_i} e^{\boldsymbol{a}_{ik}^T \boldsymbol{y} + b_{ik}} \right) \leq 0, \quad i = 1, \ldots, m \\
& \boldsymbol{c}_j^T \boldsymbol{y} + d_j = 0, \quad j = 1, \ldots, r
\end{aligned}$$

This is convex by the convexity of log-sum-exp (soft max) functions

Geometric program (cont'd)

Example. Let $u = \log x$, $v = \log y$, $w = \log z$.

$$\begin{aligned}
\min_{x,y,z>0} \quad & x^{-1}y + xz \\
\text{s. t.} \quad & 2x^{-1} \leq 1 \\
& \frac{1}{3}x \leq 1 \\
& x^2 y^{-1/2} + 3y^{1/2}z^{-1} \leq 1 \\
& x^2 y^{-1} z^{-2} = 1
\end{aligned}$$

is equivalent to

$$\begin{aligned}
\min_{u,v,w} \quad & \log(e^{-u+v} + e^{u+w}) \\
\text{s. t.} \quad & \log 2 - u \leq 0 \\
& -\log 3 + u \leq 0 \\
& \log(e^{2u-\frac{1}{2}v} + e^{\log 3+\frac{1}{2}v-w}) \leq 0 \\
& 2u - v - w = 0
\end{aligned}$$